

PHASES OF GASTRIC SECRETION

Headings

- ▼ PHASES OF GASTRIC SECRETION
 - ▼ 1. CEPHALIC PHASE
 - A. Unconditioned Reflex
 - B. Conditioned Reflex
 - ▼ 2. GASTRIC PHASE
 - A. Nervous Mechanism
 - B. Hormonal Mechanism (Gastrin)
 - ▼ 3. INTESTINAL PHASE
 - A. Initial Stage (Stimulatory)
 - B. Later Stages (Inhibitory)
 - 4. INTERDIGESTIVE PHASE
 - ▼ SCHEMATIC SUMMARY
 - PHASE COMPARISON BREAKDOWN

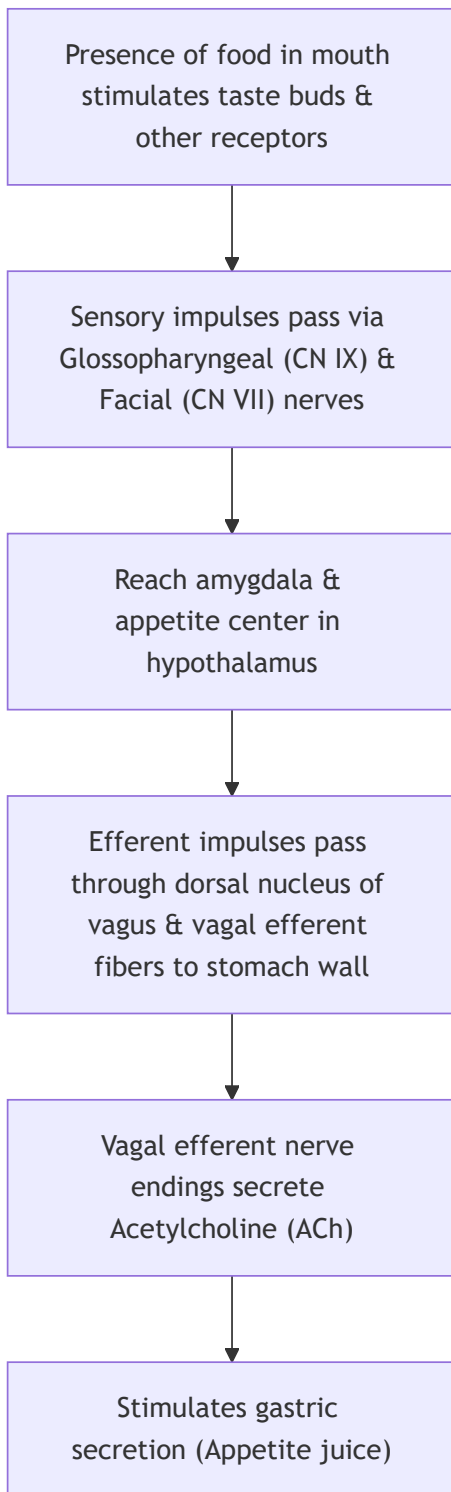
- **Nature of Secretion** : Secretion of gastric juice is a continuous process.
- **4 Phases of Gastric Secretion** :—
 - a. **Cephalic phase** (Only nervous)
 - b. **Gastric phase** (Nervous and hormonal)
 - c. **Intestinal phase** (Mostly hormonal)
 - d. **Interdigestive phase** (In humans during fasting)

1. CEPHALIC PHASE

- **Definition** : Secretion of gastric juice elicited by stimuli arising from the head region (*cephalus*).
- **Regulation** : Purely regulated by **nervous mechanism** through reflex action.
- **Appetite Juice** : Gastric juice secreted during this phase is called **appetite juice** (*quantity is less, but rich in enzymes and HCl*).

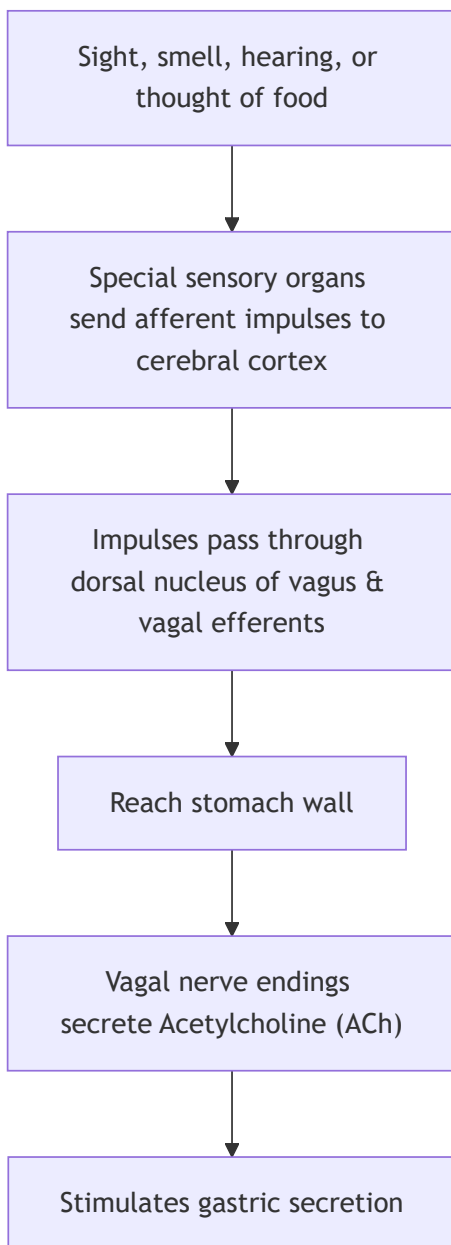
A. Unconditioned Reflex

- **Mechanism** : Inborn reflex elicited by food in the oral cavity.



B. Conditioned Reflex

- **Mechanism** : Acquired reflex from prior experience (sight, smell, hearing, or thought of food).

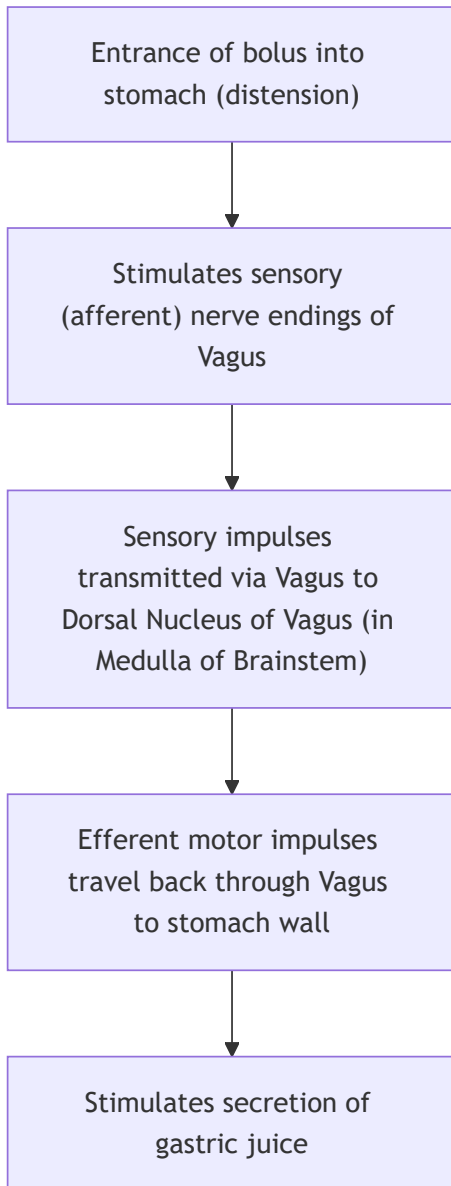


2. GASTRIC PHASE

- **Definition** : Secretion of gastric juice when food enters the stomach.
- **Regulation** : Regulated by both **nervous** and **hormonal** mechanisms.
- **Properties** : Secretes large volumes of gastric juice rich in **pepsinogen and HCl**.
- **Initiating Stimuli** :—
 - a. Distension of stomach
 - b. Mechanical stimulation of gastric mucosa (by bulk of food)
 - c. Chemical stimulation of gastric mucosa (by food contents/peptides)

A. Nervous Mechanism

- **1. Local Myenteric Reflex :—**
- Food particles stimulate the local myenteric (Auerbach's) nerve plexus in the stomach wall.
- Nerve fibers release **Acetylcholine (ACh)** → stimulates gastric glands to secrete large volumes of juice.
- *Simultaneously*, acetylcholine stimulates **G-cells to secrete Gastrin**.
- **2. Vagovagal Reflex :—**



B. Hormonal Mechanism (Gastrin)

- **Features of Gastrin :—**
- Polypeptide hormone secreted by **G-cells** in pyloric glands of stomach (*also in duodenum, jejunum, and fetal pancreatic islets*).
- Exists in forms: G_{14} , G_{17} , and G_{34} amino acids.

- **Release Mechanism :**

Local nervous reflex / Vagovagal reflex



Vagal nerve endings release **Gastrin-Releasing Peptide (GRP)**



Stimulates G-cells to secrete **Gastrin** into circulation

- **Action :** Stimulates parietal and chief cells to secrete **HCl and pepsinogen**.

3. INTESTINAL PHASE

- **Definition :** Secretion of gastric juice when chyme enters the small intestine.
- **Pattern :** Initially, gastric secretion **increases**, but later it is **inhibited/stopped**.
- **Regulation :** Regulated by nervous and hormonal control.

A. Initial Stage (Stimulatory)

Chyme entering intestine → Stimulates duodenal mucosa to release Gastrin $\xrightarrow{\text{Blood}}$ ↑ Gastric secretion

B. Later Stages (Inhibitory)

- Gastric secretion is inhibited by **two mechanisms**:
- **1. Enterogastric Reflex :—**
 - Distension or chemical/osmotic irritation of intestinal mucosa by chyme.
 - Mediated by **myenteric plexus** and **vagus nerve** → **inhibits gastric secretion & motility**.
- **2. Gastrointestinal Hormones (Inhibitory) :—**
 - a. **Gastric Inhibitory Peptide (GIP)** : Secreted in response to glucose & fats.
 - b. **Secretin** : Secreted in response to acid chyme.
 - c. **Cholecystokinin (CCK-PZ)** : Secreted in response to fats & amino acids.
 - d. **Vasoactive Intestinal Polypeptide (VIP)** : Secreted in response to acidic chyme.
 - e. **Peptide YY** : Secreted in response to fatty chyme.
 - f. **Somatostatin** : Secreted by pancreatic/intestinal D-cells → strong inhibitor of gastric secretion.

4. INTERDIGESTIVE PHASE

- **Definition** : Secretion of gastric juice in between meals or during prolonged periods of fasting.
- **Regulation** : Mainly basal secretion controlled by low circulating basal levels of hormones like **Gastrin**.

Effect of Stimulants :—

Alcohol and Caffeine significantly ↑ increase gastric acid secretion by **directly stimulating the gastric mucosa**.

SCHEMATIC SUMMARY

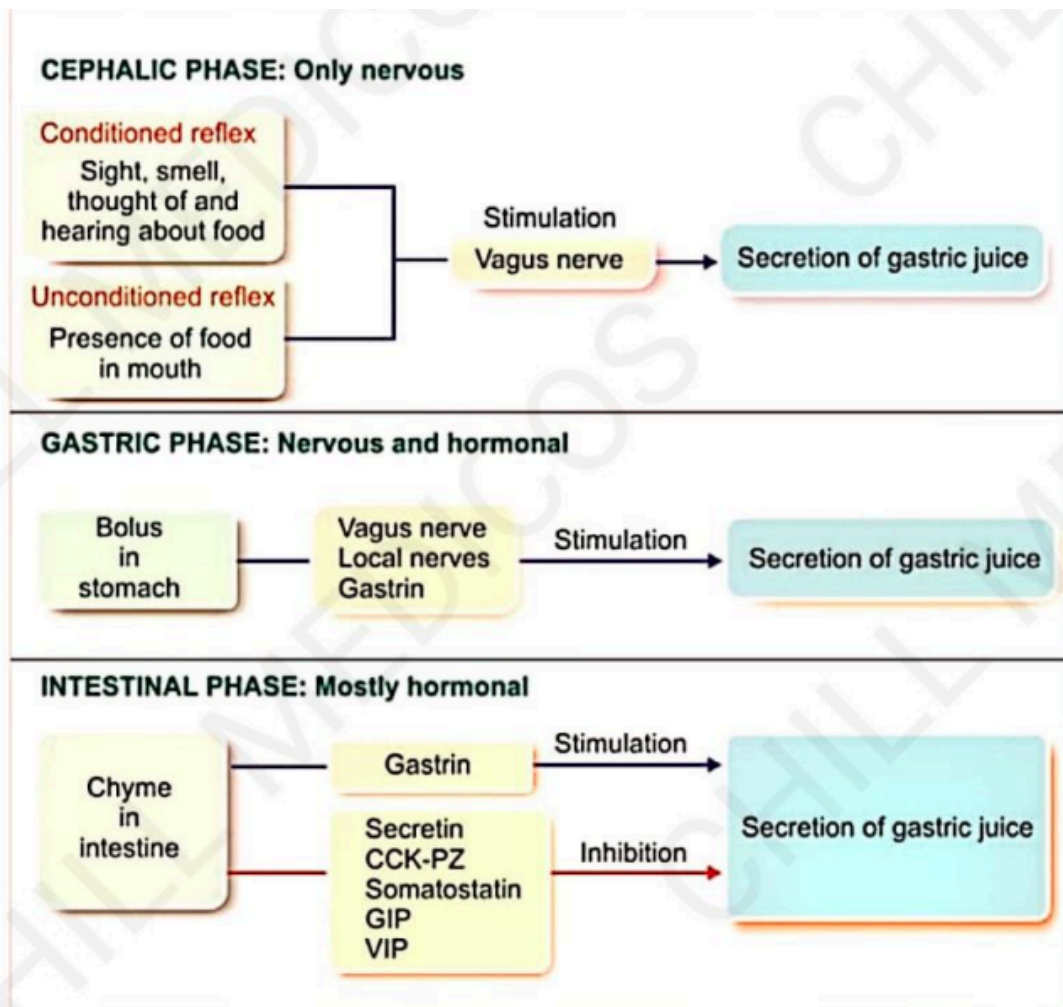
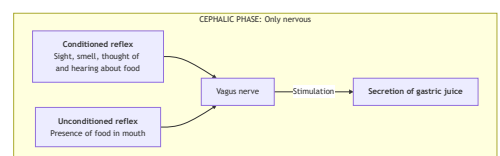
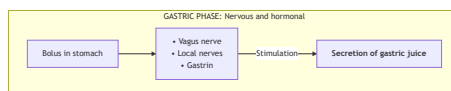
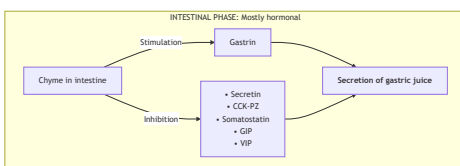


FIGURE 38.6: Schematic diagram showing the regulation of gastric secretion
CCK-PZ = Cholecystikinin-pancreozymin, GIP = Gastric inhibitory peptide, VIP = Vasoactive intestinal peptide.



PHASE COMPARISON BREAKDOWN

Key Abbreviations :—

- **CCK-PZ** : Cholecystokinin-pancreozymin
- **GIP** : Gastric inhibitory peptide
- **VIP** : Vasoactive intestinal peptide